

Bronchiectasis — Diagnosis, HRCT & Workup



1. #1 autoimmune disease

WHAT YOU SEE

Chalkboard heading '1 AUTOIMMUNE DISEASE' with a green check mark — the 'top of the class' tick marks rheumatoid arthritis as the #1 connective-tissue cause of bronchiectasis.

WHAT IT ENCODES

Rheumatoid arthritis is the most common AUTOIMMUNE/connective-tissue cause of bronchiectasis, and it can predate or follow the joint disease. The mechanism combines immune-mediated airway inflammation with susceptibility to infection. Always screen for joint symptoms, RF, and anti-CCP in otherwise-unexplained bronchiectasis (other associations: Sjögren, IBD).

BOARD TRAP

Don't assume autoimmune bronchiectasis means lupus — RA is the #1 connective-tissue association; picking SLE over RA is the discriminator trap.

2. Lymphoma risk

WHAT YOU SEE

Chalkboard line 'LYMPHOMA RISK' beside a crossed-out cell cluster — written under the autoimmune heading to flag the lymphoma risk of CVID-associated bronchiectasis.

WHAT IT ENCODES

Chronic immune dysregulation underlying bronchiectasis — particularly common variable immunodeficiency (CVID) — carries an elevated risk of lymphoma (and other malignancies/autoimmunity). This is another reason to actively identify an immunodeficiency cause rather than treating infections symptomatically.

BOARD TRAP

Recurrent infections + bronchiectasis with new lymphadenopathy/cytopenias should raise CVID-associated lymphoma — dismissing nodes as 'reactive' without considering the immunodeficiency link is the miss.

3. Black cat with medal

WHAT YOU SEE

A black cat sitting center-desk wearing a gold yin-yang medallion — the award-wearing 'best mimic' mascot for chronic bronchiectasis that masquerades for years as recurrent bronchitis/asthma.

WHAT IT ENCODES

Mascot for the chronic, recurrent, 'award-winning best mimic' nature of bronchiectasis. The defining clinical hallmark is a CHRONIC daily productive cough with copious mucopurulent sputum punctuated by exacerbations — not a single acute illness. It is frequently mislabeled as 'recurrent bronchitis' or refractory asthma/COPD for years before HRCT reveals it.

BOARD TRAP

Treating chronic productive cough as repeated episodes of acute bronchitis (and just giving antibiotic courses) misses the underlying structural diagnosis — the chronicity between exacerbations is the tell.

4. Tram-track / railway

WHAT YOU SEE

A length of toy railway track labelled TRAM-TRACKS running across the desk — the parallel rails ARE the tram-track sign: thickened, non-tapering bronchial walls seen lengthwise.

WHAT IT ENCODES

The tram-track sign = parallel, non-tapering, thickened bronchial walls seen LENGTHWISE (the dilated airway runs horizontally in the imaging plane). It appears on both chest X-ray and HRCT and reflects the same dilated, thick-walled airways that produce the signet-ring sign in cross-section.

BOARD TRAP

Tram-tracks are a sign of dilated thick-walled BRONCHI (bronchiectasis), not the Kerley B lines/interstitial markings of pulmonary edema or the reticulation of fibrosis — misreading them as edema/fibrosis is the imaging trap.

5. Signet-ring (ring + diamond)

WHAT YOU SEE

An open jewelry box holding a diamond ring tagged 'SIGNET RING $\geq 1.5\times$ ' — the ring (dilated bronchus) beside its tiny diamond (the smaller artery) IS the signet-ring sign, broncho-arterial ratio >1 .

WHAT IT ENCODES

The signet-ring sign is the diagnostic HRCT hallmark: a dilated bronchus (the ring) sits adjacent to its smaller accompanying pulmonary artery (the 'stone'), giving a broncho-arterial ratio >1 (roughly ≥ 1.5). Seen in CROSS-section, it is the cardinal CT criterion for bronchiectasis.

BOARD TRAP

Normally the bronchus is SMALLER than its paired artery — so a bronchus LARGER than the adjacent artery (BA ratio >1) defines bronchiectasis; reversing the normal relationship is the tested point.

6. Tree-in-bud

WHAT YOU SEE

A small potted bare tree labelled TREE-IN-BUD — the budding branches depict centrilobular tree-in-bud nodules of active small-airway infection (think NTM).

WHAT IT ENCODES

Tree-in-bud opacities are centrilobular nodules with branching lines representing mucus/pus/inflammation filling small airways. They signal ACTIVE small-airway infection or inflammation — classically MAC/NTM, but also any infective exacerbation or aspiration — rather than the structural dilation itself.

BOARD TRAP

Tree-in-bud indicates active airway INFECTION (think NTM, infectious exacerbation), not chronic structural change — if present, pursue sputum cultures including AFB rather than assuming inert scarring.

7. CT >1 criteria card

WHAT YOU SEE

A pinned index card reading 'CT ≥ 1 : signet ring / no tapering / periph airways' — lists the three HRCT diagnostic features of bronchiectasis.

WHAT IT ENCODES

The three core HRCT diagnostic features: (1) broncho-arterial ratio >1 (signet ring), (2) lack of normal distal bronchial TAPERING, and (3) bronchi VISIBLE in the peripheral lung (within ~ 1 cm of the pleura). HRCT is the gold-standard confirmatory test for bronchiectasis.

BOARD TRAP

Seeing airways within ~ 1 cm of the pleura is ABNORMAL — peripheral bronchi shouldn't normally be visible, so calling visible peripheral airways 'normal' misses the diagnosis.

8. Clinical card (cough/sputum)

WHAT YOU SEE

A pinned index card reading 'CLINICAL ≥2: cough / sputum / exacerbations' — the clinical triad that should prompt confirmatory HRCT.

WHAT IT ENCODES

Clinical diagnosis rests on chronic productive cough with copious purulent/mucopurulent sputum, recurrent exacerbations, and sometimes hemoptysis; exam may reveal crackles and clubbing. This clinical picture should prompt confirmatory HRCT, and chronic *Pseudomonas* colonization predicts worse outcomes.

BOARD TRAP

A normal chest X-ray does NOT exclude bronchiectasis — if the clinical picture fits, get HRCT; relying on plain film alone under-diagnoses.

9. IgE thermometer (ABPA)

WHAT YOU SEE

A rising thermometer labelled IgE beside a card 'REFRACTORY ASTHMA → ABPA' — the sky-high IgE thermometer in a refractory asthmatic points to ABPA.

WHAT IT ENCODES

A very high total IgE (often >1000 IU/mL) in an asthmatic — especially with REFRACTORY/poorly controlled asthma and central bronchiectasis — should trigger an ABPA workup: *Aspergillus*-specific IgE/IgG, peripheral eosinophilia, and serum precipitins. Treatment is corticosteroids ± itraconazole.

BOARD TRAP

A high IgE in an asthmatic with new bronchiectasis is ABPA until proven otherwise — simply escalating inhaled asthma medication without checking *Aspergillus* serologies misses (and undertreats) ABPA.

10. CF FIRST jar

WHAT YOU SEE

A top-shelf jar labelled CF FIRST sitting on a gold star — the gold star marks cystic fibrosis (sweat chloride/genetics) as the first test to order in a young patient.

WHAT IT ENCODES

In a young patient (especially with upper-lobe disease, sinus disease, malabsorption, or infertility), test for cystic fibrosis FIRST — sweat chloride testing (>60 mmol/L abnormal) and CFTR genetic analysis. CF is high-yield because it is common, heritable, and changes management dramatically.

BOARD TRAP

Skipping CF testing in a young adult with bronchiectasis because they 'would have been diagnosed in childhood' is wrong — mild CFTR variants can present in adulthood, so sweat chloride/genetics stay first-line.

11. IgG / immunodeficiency jar

WHAT YOU SEE

A shelf jar of antibodies labelled IgG / IgA / IgM — represents quantitative immunoglobulin testing to catch CVID/hypogammaglobulinemia.

WHAT IT ENCODES

Measure serum immunoglobulins (IgG, IgA, IgM), and consider IgG subclasses and specific antibody/vaccine responses, to detect CVID, hypogammaglobulinemia, or selective IgA deficiency as causes of recurrent infection-driven bronchiectasis. Identifying these enables immunoglobulin replacement.

BOARD TRAP

Recurrent sinopulmonary infections with bronchiectasis warrant quantitative immunoglobulins — treating each infection in isolation and never measuring Ig levels misses a correctable immunodeficiency.

12. AAT (alpha-1) jar

WHAT YOU SEE

A lower-shelf jar labelled AAT — alpha-1 antitrypsin testing, sought in early-onset basal disease, especially a nonsmoker.

WHAT IT ENCODES

Alpha-1 antitrypsin deficiency is a less common but testable cause; the unopposed neutrophil elastase classically produces BASAL panacinar EMPHYSEMA, and bronchiectasis may coexist. Check serum AAT level (and phenotype/genotype if low), particularly with early-onset basal disease or liver involvement.

BOARD TRAP

AAT deficiency classically causes LOWER-zone emphysema (vs the upper-zone disease of smoking-related COPD) — overlooking AAT testing in early-onset basal disease, especially a nonsmoker, is the workup trap.

13. Cilia biopsy

WHAT YOU SEE

A rainbow pinwheel/whistle on a cord tagged CILIA BIOPSY at the bottom — the spinning cilia toy = ciliary ultrastructure biopsy (with nasal NO) to confirm primary ciliary dyskinesia.

WHAT IT ENCODES

For suspected primary ciliary dyskinesia (bronchiectasis + chronic sinusitis ± situs inversus ± infertility), screen with low nasal nitric oxide and confirm with ciliary ULTRASTRUCTURE on biopsy (electron microscopy showing dynein-arm defects) and/or genetic testing.

BOARD TRAP

Don't confirm PCD with a sweat chloride (that's CF) — the right confirmatory tests are nasal NO and ciliary biopsy/genetics; using CF testing for a PCD picture is the mismatch trap.

14. IVIG drip (Q3-4 weeks)

WHAT YOU SEE

An IV pole on the far left hanging a bag labelled 'IVIG Q3-4 WEEKS' (with an 'SC weekly' tag) — immunoglobulin replacement reserved for the antibody-deficient subgroup.

WHAT IT ENCODES

Immunoglobulin replacement (IVIG, typically every 3–4 weeks, or weekly SCIG) is the treatment for the ANTIBODY-DEFICIENCY subgroup (CVID, agammaglobulinemia, significant hypogammaglobulinemia) and reduces the frequency of infections and exacerbations.

BOARD TRAP

IVIG is ONLY for the immunodeficient subgroup — it is NOT a general bronchiectasis therapy; prescribing it for idiopathic or post-infectious bronchiectasis with normal immunoglobulins is wrong.

15. Chronic not acute

WHAT YOU SEE

A tag reading 'CHRONIC - NOT JUST ACUTE' pinned on the slumped patient — underscores bronchiectasis as a chronic irreversible structural disease, not a one-off pneumonia.

WHAT IT ENCODES

Bronchiectasis is a CHRONIC, irreversible STRUCTURAL airway disease; symptoms (cough, sputum) persist between exacerbations, distinguishing it from a single resolving acute pneumonia. Management centers on long-term airway clearance, treating exacerbations, and addressing colonization (e.g., chronic macrolide for frequent exacerbators; inhaled antibiotics for Pseudomonas).

BOARD TRAP

Treating bronchiectasis as a one-off acute pneumonia — antibiotics then discharge with no airway-clearance/long-term plan — ignores its chronic structural nature and leads to recurrent admissions.

16. Mild-mod destruction (overlap)

WHAT YOU SEE

A label on the pinwheel cord reading 'MILD→MOD OBSTRUCTION OVERLAPS COPD' at the bottom-center — points to the cylindrical-to-cystic severity spectrum that overlaps clinically with COPD.

WHAT IT ENCODES

Bronchiectasis severity grades morphologically from cylindrical (tubular, mild) to varicose (beaded, moderate) to cystic/saccular (severe), and these patterns often OVERLAP within the same lung. Severity (and Pseudomonas colonization) correlates with exacerbation frequency and prognosis, captured by indices like the Bronchiectasis Severity Index.

BOARD TRAP

Cylindrical (mild) bronchiectasis is still real, treatable disease — dismissing milder cylindrical changes as insignificant because they aren't cystic underestimates exacerbation risk.